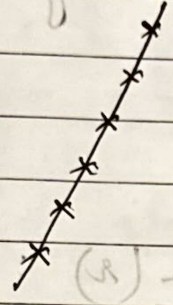
CORRELATION & REGRESSION

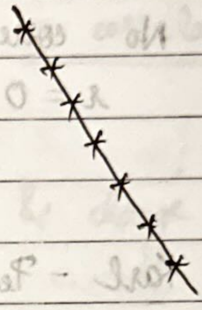
→ Coeff of correlation = $\underline{\underline{r}}$: $-1 \le r \le 1$

* **Types of Correlation** :

1.

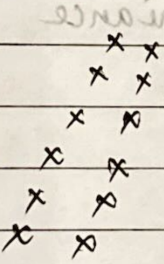


perfect +ve correlation
$r = +1$

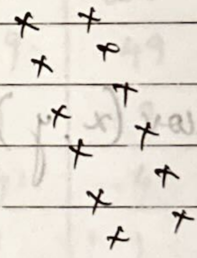


perfect −ve correlation
$r = -1$

2.

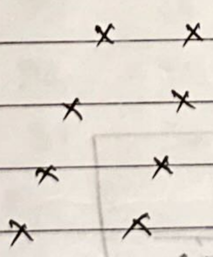


High deg +ve correlation
r : +ve close to 1

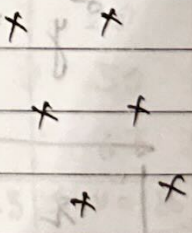


High deg −ve correlation
r : −ve close to −1

3.



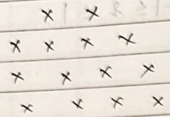
Low deg +ve correlation
r : +ve away from 1



Low deg −ve correlation
r : −ve away from −1

CORRELATION & REGRESSION

4.



No correlation

$r = 0$

$\bar{x} \Rightarrow$ mean of x

$\bar{y} \Rightarrow$ mean of y

$N \Rightarrow$ no. of values

$x_i \Rightarrow$ value of x

$y_i \Rightarrow$ value of y

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* Karl - Pearson's Co-eff of correlation (r) :

- $$\boxed{r = \frac{cov(x\,y)}{\sigma_x \times \sigma_y}}$$

σ_x : std deviation of x

σ_y : std deviation of y

cov : Covariance

$$cov(x,y) = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{N}$$

$$\sigma_x = \sqrt{\frac{\Sigma (x_i - \bar{x})^2}{N}}$$

$$\sigma_y = \sqrt{\frac{\Sigma (y_i - \bar{y})^2}{N}}$$

- $$\boxed{r = \frac{\Sigma (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\Sigma (x_i - \bar{x})^2}\sqrt{\Sigma (y_i - \bar{y})^2}}}$$

- Also, $$\boxed{r = \frac{\Sigma x_i y_i - N\bar{x}\bar{y}}{\sqrt{\Sigma x_i^2 - N\bar{x}^2}\sqrt{\Sigma y_i^2 - N\bar{y}^2}}}$$

Note :

- r range : $-1 \le r \le 1$
- if x & y are independent, there is no correlation b/w x & y (spurious correlation)
- r is independent of change of origin & change of scale

Q. Calculate r for following data :

x	y	$u = x-30$	$v = y-25$	u^2	v^2	uv
23	18	-7	-7	49	49	49
27	22	-3	-3	9	9	9
28	23	-2	-2	4	4	4
29	24	-1	-1	1	1	1
30	25	0	0	0	0	0
31	26	1	1	1	1	1
33	28	3	3	9	9	9
35	29	5	4	25	16	20
36	30	6	5	36	25	30
39	32	9	7	81	49	63
N = 10		$\Sigma u = 11$	$\Sigma v = 7$	$\Sigma u^2 = 215$	$\Sigma v^2 = 163$	$\Sigma uv = 186$

$$\bar{u} = \frac{11}{10} = 1.1$$

$$\bar{v} = \frac{7}{10} = 0.7$$

\# $r = \dfrac{\sum x_i y_i - N\bar{x}\bar{y}}{\sqrt{\sum x_i^2 - N\bar{x}^2}\sqrt{\sum y_i^2 - N\bar{y}^2}}$

$= \dfrac{\sum uv - N\bar{u}\bar{v}}{\sqrt{\sum u^2 - N\bar{u}^2}\sqrt{\sum v^2 - N\bar{v}^2}} = \dfrac{186 - 10(1.1)(0.7)}{\sqrt{208.9}\sqrt{158.1}}$

$\boxed{r = 0.995}$ ⇒ High degree +ve correlation

* Spearman's Rank Correlation Coefficient (R)

- $\boxed{R = 1 - \left[\dfrac{6 \sum d_i^2}{n^3 - n}\right]}$ $\sum d_i$: diffⁿᶜᵉ b/w 2 ranks, n : no. of obsr.

- $-1 \le R \le 1$

Q. Calc. R from following :

S.No	Rank in Eng (R_1)	Rank in Maths (R_2)	$d^2 = (R_1 - R_2)^2$
1	1	3	4
2	3	1	4
3	7	4	9
4	5	5	0
5	4	6	4
6	6	9	9
7	2	7	25
8	10	8	4
9	9	10	1
10	8	2	36

$\sum d^2 = 96$

$R = 1 - \dfrac{6(96)}{990}$

$= 1 - 0.58$

$= 0.42$

$\boxed{R = 0.42}$

\# $\sum d \Rightarrow$ always zero

11/1/24

$R = 1 - \dfrac{6 \sum d^2}{n^3 - n}$

→ For equal ranks

$R = 1 - \dfrac{6\left[\sum d^2 + \frac{1}{12}(m_1^3 - m_1) + \frac{1}{12}(m_2^3 - m_2) + \dots\right]}{n^3 - n}$

where m_i = no. of times item is repeated

Q. Calculate R for the following data :

X	Y	R_1	R_2	d	d^2
52	40	3	5	2	4
55	30	9	3.5	5.5	30.25
49	70	7.5	9	1.5	2.25
60	20	10	1	9	81
43	30	5.5	3.5	2	4
37	50	4	7	3	9
43	72	5.5	10	4.5	20.25
49	60	7.5	8	0.5	0.25
10	45	1	6	5	25
20	25	2	2	0	0
					176

$n = 10$

$m_1 = 2$ (for 43)

$m_2 = 2$ (for 49)

$m_3 = 2$ (for 30)

$d = R_1 - R_2$

$R = 1 - \dfrac{6\left[176 + \frac{1}{12}(8-2) + \frac{1}{12}(8-2) + \frac{1}{12}(8-2)\right]}{1000 - 100}$

$$= 1 - \frac{6(176 + 1.5)}{990}$$

$$= 1 - 1.075$$
$$= -0.075$$

Q. $r = 0.4$
$cov(x,y) = 1.6$
$\sigma_y^2 = 25$ find σ_x

$$r = \frac{cov(x,y)}{\sigma_x \times \sigma_y}$$

$$\sigma_x \times \sigma_y = \frac{cov(x,y)}{r}$$

$$\sigma_x = \frac{1.6}{0.4 \times 5} = 0.8$$

Q. $R = 0.143$
$\Sigma d^2 = 48$

$$R = 1 - \frac{6 \Sigma d^2}{n^3 - n}$$

$$R + \frac{6 \Sigma d^2}{n^3 - n} = 1$$

$$0.143 n^3 - 0.143 n + 48 = n^3 - n$$
$$0.857 n + 48 = 0.857 n^3$$
$$-0.857 n^3 + 0.857 n + 48 = 0$$

$$n^3 - n = 336.056$$
$$n = 7.0003$$
$$n = 7$$

Q. Find r for $N = 10$, $\Sigma x = 140$, $\Sigma y = 150$, $\Sigma(x-10)^2 = 180$
$\Sigma(y-15)^2 = 215$, $\Sigma(x-10)(y-15) = 60$

$$r_{uv} = \frac{\Sigma uv - n\bar{u}\bar{v}}{\sqrt{\Sigma u^2 - n\bar{u}^2}\sqrt{\Sigma v^2 - n\bar{v}^2}}$$

$$r_{xy} = \frac{\Sigma xy - n\bar{x}\bar{y}}{\sqrt{\Sigma x^2 - n\bar{x}^2}\sqrt{\Sigma y^2 - n\bar{y}^2}}$$

$u = x - a$
$v = y - b$

Let $u = x - 10$
$v = y - 15$

$\Sigma u^2 = 180$
$\Sigma v^2 = 215$
$\Sigma uv = 60$

$\Sigma u = \Sigma(x - 10)$
$= \Sigma x - \Sigma 10$
$= 140 - 10 \times 10$
$= 40$

$\Sigma v = \Sigma(y - 15)$
$= \Sigma y - \Sigma 15$
$= 150 - 15 \times 10$
$= 0$

$\therefore \bar{u} = \frac{\Sigma u}{N} = \frac{40}{10} = 4$

$\bar{v} = \frac{\Sigma v}{N} = \frac{0}{10} = 0$

$$\therefore r_{uv} = \frac{60 - 10\,(4\times 0)}{\sqrt{180-160}\,\sqrt{215-0}}$$

$$= \frac{60}{\sqrt{20}\,\sqrt{215}} = \frac{60}{65.574}$$

$$\boxed{r_{uv} = 0.915}$$

Q. 10 competetors in a musical test were ranked by 3 judges x, y, z in the following order:
Using rank correlation method, discuss which pair of judges has the nearest approach to the common likings in music.

R_1 x	R_2 y	R_3 z	R_1-R_2 d_{xy}	R_2-R_3 d_{yz}	R_1-R_3 d_{xz}	d_{xy}^2	d_{yz}^2	d_{xz}^2
1	3	6	2	3	5	4	9	25
6	5	4	1	1	2	1	1	4
5	8	9	3	1	4	9	1	16
10	4	8	6	4	2	36	16	4
3	7	1	4	6	2	16	36	4
2	10	2	8	8	0	64	64	0
4	2	3	2	1	1	4	1	1
9	1	10	8	9	1	64	81	1
7	6	5	1	1	2	1	1	4
8	9	7	1	2	1	1	4	1
					Σ	200	214	60

$$R_{xy} = 1 - \frac{6(200)}{990} = 1 - 1.211 = -0.212$$

$$R_{yz} = 1 - \frac{6(214)}{990} = 1 - 1.296 = -0.296$$

$$R_{xz} = 1 - \frac{6(60)}{990} = 1 - 0.364 = 0.636$$

+ve

$\therefore$ judge x & z have similar likings in music.

12/1/24

Q. A sample of 25 pairs of (x,y) values gives the following results

$\Sigma x = 127$ $\Sigma x^2 = 760$
$\Sigma y = 100$ $\Sigma y^2 = 449$
$\Sigma xy = 500$

Later on it was found that 2 pairs of values were taken as (8, 14) & (8, 6) instead of (8, 12) & (6, 8). Find correct correlation coefficient.

$$\text{Correct } \Sigma x = \text{wrong } \Sigma x - \underbrace{(8+8)}_{\text{wrong}} + \underbrace{(8+6)}_{\text{correct}}$$
$$= 127 - 16 + 14 = 125$$

$$\text{Corret } \Sigma y = \text{wrong } \Sigma y - (14+6) + (12+8)$$
$$= 100$$

$$\text{Correct } \Sigma x^2 = \text{wrong } \Sigma x^2 - (8^2+8^2) + (8^2+6^2)$$
$$= 760 - 128 + 100 = 732$$

$$\text{Correct } \Sigma y^2 = \text{wrong } \Sigma y^2 - (14^2+6^2) + (12^2+8^2)$$
$$= 449 - (14^2+6^2) + (12^2+8^2)$$
$$= 425$$

$$\text{Correct } \Sigma xy = \text{wrong } \Sigma xy - (8\times14 + 8\times6) + (8\times12 + 6\times8)$$
$$= 500 -$$
$$= 484$$

$$\therefore \text{ Correct } \bar{x} = \frac{\Sigma x}{n} = \frac{125}{25} = 5$$

$$\therefore \text{ Correct } \bar{y} = \frac{\Sigma y}{n} = \frac{100}{25} = 4$$

$$r = \frac{\Sigma xy - n\bar{x}\bar{y}}{\sqrt{\Sigma x^2 - n\bar{x}^2}\sqrt{\Sigma y^2 - n\bar{y}^2}}$$

$$= \frac{484 - 25(20)}{\sqrt{732-625}\sqrt{425-400}}$$

$$= \frac{-16}{10.34 \times 5}$$

$$r = \underline{-0.309}$$

* **Regression Lines:**

1. Regression line of y on x

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

where $b_{yx} = r\frac{\sigma_y}{\sigma_x}$ (Reg coeff)

Teacher's Signature:..................

2. Reg line of x on y

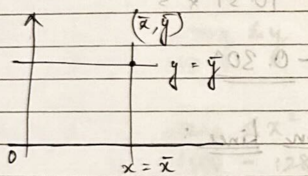
$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$b_{xy} = r\frac{\sigma_x}{\sigma_y}$$

→ b_{yx} & b_{xy} are the slopes on the regression lines of y on x & x on y respectively

Remarks

① if $r = 0$, then $x = \bar{x}$ & $y = \bar{y}$



if $r = \pm 1$, then both lines are parallel to each other

② $$b_{yx} = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2}$$

$$b_{xy} = \frac{\sum xy - n\bar{x}\bar{y}}{\sum y^2 - n\bar{y}^2}$$

③ $r = \pm\sqrt{b_{xy}\, b_{yx}}$

④ If $b_{yx} > 1$ then $b_{xy} < 1$

⑤ $\frac{b_{xy} + b_{yx}}{2} \geq r$

⑥ If θ is the angle b/w 2 reg. lines then

$$\tan\theta = \left(\frac{1 - r^2}{r}\right)\frac{\sigma_x \times \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

⇓

$$\boxed{\tan\theta = \left(\frac{1 - r^2}{r}\right)\frac{\sigma_x \times \sigma_y}{\sigma_x^2 + \sigma_y^2}}$$

→ if $r = 0 \Rightarrow \tan\theta = \infty \Rightarrow \theta = \pi/2$ ⊥ li

→ if $r = \pm 1 \Rightarrow \tan\theta = 0 \Rightarrow \theta = 0$ || li

⑦ Reg coeff are independent of change of origin but not change of scale.

if $u = \frac{x - a}{h}$ $\quad v = \frac{y - b}{k}$

then

$$\sigma_x = h\,\sigma_u$$
$$\sigma_y = k\,\sigma_v$$

$$\therefore b_{yx} = \frac{r\sigma_y}{\sigma_x} = \frac{k}{h}\frac{r\,\sigma_v}{\sigma_u}$$

$$= \frac{k}{h}\, b_{vu}$$

$$b_{xy} = \frac{h}{k}\, b_{uv}$$

Q. A panel of 2 judges A & B graded the following performances

ans part

Performance No.	A (x)	B (y)	u $(x-\bar{x})$	v $(y-\bar{y})$	uv	u^2
1	36	35	3	2	6	9
2	32	33	-1	0	0	1
3	34	31	1	-2	-2	1
4	31	30	-2	-3	6	4
5	32	34	-1	1	-1	1
6	32	32	-1	-1	1	1
7	34	36	1	3	3	1
	$\Sigma x = 231$	$\Sigma y = 231$	$\Sigma u = 0$	$\Sigma v = 0$	$\Sigma uv = 13$	$\Sigma u^2 = 18$

Judge A awarded 38 marks to 8^{th} performance & B was absent. If B was present, how many marks would he have awarded?

→ $\bar{u} = \frac{\Sigma u}{n} = 0$

$\bar{v} = \frac{\Sigma v}{n} = 0$

$v - \bar{v} = b_{vu}(u - \bar{u})$

$b_{vu} = \frac{\Sigma uv - n\bar{u}\bar{v}}{\Sigma u^2 - n\bar{u}^2}$

$= \frac{13}{18} = 0.72$

$\therefore v - \bar{v} = b_{vu}(u - \bar{u})$

$\Rightarrow v = 0.72\,u \quad (\because \bar{u} = \bar{v} = 0)$

$(y - \bar{y}) = 0.72(x - \bar{x})$

$\Rightarrow y = 33 + 0.72(x - 33)$

when $x = 38$

$\therefore y = 33 + 0.72(38 - 33)$

$= 36.6$

$y \approx 37$ ans.

Q. Following is the information about 60 students

	Maths	Eng
Mean	80 ($\bar{x}$)	50 ($\bar{y}$)
S.D	15 (σ_x)	10 (σ_y)

$r = 0.4$

Estimate marks of student in maths who scored 60 marks in Eng. & marks of a student in Eng who scored 70 in maths.

→ Maths(x) Eng(y)

→ $y - \bar{y} = b_{yx}(x - \bar{x})$

$x = 70$, $y = ?$

→ $x - \bar{x} = b_{xy}(y - \bar{y})$

$y = 60$, $x = ?$

→ $b_{yx} = r\dfrac{\sigma_y}{\sigma_x}$ $\qquad b_{xy} = r\dfrac{\sigma_x}{\sigma_y}$

$= 0.4 \times \dfrac{10}{15}$ $\qquad = 0.4 \times \dfrac{15}{10}$

$= 0.267$ $\qquad = 0.6$

→ $y - \bar{y} = b_{yx}(x - \bar{x})$

$y = 0.267(70 - 80) + 50$

$= 47.33 \approx 47$

$x - \bar{x} = b_{xy}(y - \bar{y})$

$x = 0.6(60 - 50) + 80$

$= 86$

Q. Given that $6y = 5x + 90$

$15x = 8y + 130$

$\sigma_x^2 = 16$

find ① $\bar{x}$ & $\bar{y}$

② r

③ σ_y^2

→ $15x - 8y = 130$

$3(-5x + 6y = 90)$

$\cancel{15x} - 8y = 130$

$-\cancel{15x} + 18y = 270$

$10y = 400$

$\bar{y} = 40$

$-5x = 90 - 240$

$\bar{x} = \dfrac{-150}{-5} = 30$

① $\bar{x} = 30$ $\quad \bar{y} = 40$

② (i) ⟹ $y = \dfrac{5}{6}x + \dfrac{90}{6} \Rightarrow b_{yx} = \dfrac{5}{6}$

(2) ⟹ $x = \dfrac{8}{15}y + \dfrac{130}{15} \Rightarrow b_{xy} = \dfrac{8}{15}$

$r = \sqrt{b_{xy}\, b_{yx}}$

$2/3$

③ $b_{xy} = r\dfrac{\sigma_x}{\sigma_y}$

$\sigma_y = \dfrac{2}{3} \cdot \dfrac{4}{8} \cdot 15$

$\sigma_y^2 = 25$

Teacher's Signature: ....

Teacher's Signature: ....

Q. If $\sigma_x = \sigma_y = \sigma$ & $\theta = \tan^{-1}(3)$
find r

→ $\tan\theta = \left(\frac{1-r^2}{r}\right)\left(\frac{\sigma_x \times \sigma_y}{\sigma_x^2 + \sigma_y^2}\right)$

$= \frac{1-r^2}{r} \times \frac{\sigma^2}{2\sigma^2}$

$\tan\theta = \frac{1-r^2}{2r}$

$3 = \frac{1-r^2}{2r}$

$1-r^2 = 6r$

$r^2+6r-1=0$

$r = -3 \pm \sqrt{10}$

$= -3+\sqrt{10}$

$\boxed{r = 0.16}$

Q. If $\bar{x} = 5$ $\bar{y} = 10$ & line of regression of y on x is $\parallel$ to the line $20y = 9x + 40$
Find y for $x = 30$

→ $20y = 9x + 40$

$y = \frac{9}{20}x + 2$

$b_{yx} = \frac{9}{20}$

$y - \bar{y} = b_{yx}(x - \bar{x})$

$y - 10 = \frac{9}{20}(\cancel{x} - 30 - 5)$

$y = 11.25 + 10$

$\boxed{y = 21.25}$

Q. If tangent of the angle made by line of reg of y on x is 0.6 and $\sigma_y = 2\sigma_x$
Find r

→ ~~$\tan\theta = 0.6$~~

~~$\sigma_y = 2\sigma_x$~~

~~$\tan\theta = \left(\frac{1-r^2}{r}\right)\left(\frac{\sigma_x \times \sigma_y}{\sigma_x^2 + \sigma_y^2}\right)$~~

~~$0.6 = \left(\frac{1-r^2}{r}\right)\left(\frac{2\sigma_x^2}{5\sigma_x^2}\right)$~~

~~$\frac{5}{2} \times 0.6 = \left(\frac{1-r^2}{r}\right)$~~

~~$r^2 + 1.5r - 1 = 0$~~

~~$r = 0.5 \quad -2$~~

~~$r = 0.5$~~

→ $b_{yx} = 0.6$ $\sigma_y = 2\sigma_x$

$r\frac{\sigma_y}{\sigma_x} = 0.6 \Rightarrow r\frac{2\sigma_x}{\sigma_x} = 0.6 \Rightarrow r = 0.3$